CSCI 150 | Eye-DHD Functional Requirements

Feature Name: Data Cleaning		**Priority**: HIGH
Description: The data supplied by the Psychology department contains data not necessary for this project. In order to better work with the Data for processing, it must be first cleaned of unnecessary data. If the provided CSV file does not match the correct format, it will not be cleaned and the program will display an error.		
Inputs: CSV File	**Outputs**: CSV File	**Processes**: The following columns will be removed from the entered CSV File: ... Once the specified columns above are removed from the CSV File, it will be renamed to [original_name]"CLEANED" . csv. Example: (EyeData01.csv) -> (EyeData01CLEANED.csv).
Dependencies: csvkit (csvkit is a suite of command-line tools for converting to and working with CSV, the king of tabular file formats.)		

Feature Name: Data to Animation Conversion		**Priority**: HIGH
Description: An animation shall be generated upon completion of cleaning the file to be processed. The animation will display a real-time representation of the pupils, the irises, the pupil measurements in mm, the frame number, and the timestamp. The user shall have the option to download the animation or proceed to the video editing section of the app passing the animation to be worked on.		
Inputs: Output from data cleaning step	**Outputs**: Pupil animation in mp4 format	**Processes**: The clean data will be translated into rotations on the roll, pitch, and yaw. Pupil dilation will also be reflected on the model. This animation will be used to reference
Dependencies: Data cleaning		

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Feature Name: File Upload/Download		**Priority**: MEDIUM
Description: The system shall allow users to upload raw eye-tracking data (CVS files) and experiment video files into the platform. Uploaded files shall be validated for type, size and required structure before being accepted and saved into local storage. PostgreSQL shall maintain metadata such as file paths, session associations, and processing status. After processing, the system shall provide secure download options so that users can retrieve cleaned data and generated or composited video outputs.		
Inputs: CSV files, Video Files	**Outputs**: Video Files, Clean Data	**Processes**: The User selects files to upload -> application validates type/size then writes records in PostgreSQL -> After processing completes, secure download
Dependencies: Data cleaning, Data to Animation Conversion		

What

Feature Name: Video editor		**Priority**: MEDIUM
Description: The user shall be capable of adding a video file of the simulation and a video file of the associated animation of the eyes to the video editing interface. The user shall be able to drag and drop the videos onto a canvas to display the videos simultaneously. The user shall be able to move the videos along a timeline in order to synchronize playback referring to the same frame/timestamp. Upon user satisfaction of the syncing process, the user shall be able to export the combined video to their device.		
Inputs: Video file of simulation, Video file of pupil animation	**Outputs**: Output file of the combined, synchronized video.	**Processes**: Implement video editor API and export options
Dependencies: Data cleaning, file upload, pupil animation		

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Feature Name: User Interface		**Priority**: LOW
Description: The dashboard will serve as the main interface for researchers and users to interact with EyeDHD. Researchers will be able to load their dataset into the system. From there, they can initiate data visualization and view animations of eye-tracking metrics. The interface will guide the researchers through each step with clear and easy-to-follow instructions. As a result, the researchers will be able to view and analyze visualizations and outputs for each use case.		
Inputs: CSV files and their corresponding video files	**Outputs**: Cleaned data, animations, export files	**Processes**: Manage the workflow: Load -> clean -> analyze -> review. Display the analysis results, handle any occurring errors, and allow the exporting of findings
Dependencies: Data cleaning, pupil animation, file upload/download, React (front-end), Electron (deployment), PostgreSQL(data storage)		

CSCI 150 | Eye-DHD Non-Functional Requirements

NFR	Example
Availability	If provided with internet access, the tool shall be available to use.
Compatibility	Software shall be able to be run on the operating systems, Windows 11, and built with preparation to work on future Windows operating systems. The software shall run and be installed locally, with one installation per machine (as needed). The software shall only handle one operation at a time, this means one CSV file to be processed to a video will be processed at a time.
Maintainability	This software shall be hosted on a public github repository and is open source. It shall always be available to be downloaded from this public repository for easy installation. This github shall be where updates are pushed, and new software versions must be reinstalled, not updated.
Portability	It shall be a portable application, which means it can be run from a portable storage device (like a flashdrive). The software shall not require an installation wizard/program.
Usability	The user shall be able to identify the navigation bar and where each link/button points with minimal training. The interface shall be self-explanatory, such as names for each step of the process for the tool.
Functionality	The user shall be able to import CSV data files for data cleaning and given the option to export the cleaned data file. The user shall be able to import or pass a cleaned data file for processing the data into the eye animation. The user shall be able to import/pass the eye animation and the associated video file to be combined into a single video for analysis by the department.